

Smart Research Assistant

Problem Statement No. 1 – Research Agent

The Challenge

Research and innovation require analyzing vast amounts of academic and scientific information. However, researchers, students, and professionals often spend significant time searching for relevant literature, reviewing papers, managing references, and extracting key insights from multiple sources.

Research information is scattered across journals, repositories, and databases, making knowledge discovery time-consuming and challenging. Without intelligent assistance, conducting literature reviews and generating research insights can be inefficient and resource-intensive.

The Objective

Build an AI-powered Research Agent that assists researchers and students in discovering, analyzing, and summarizing academic information efficiently.

The solution should:

- Search and retrieve relevant research papers.
- Summarize academic articles and publications.
- Perform RAG-based knowledge retrieval.
- Generate research insights and recommendations.
- Assist with literature reviews.
- Support report and documentation generation.
- Improve research productivity and accuracy.

Technology Stack

LangFlow

IBM watsonx.ai

IBM Granite Models

ChromaDB

Retrieval-Augmented Generation (RAG)

IBM Cloud Lite Services

arXiv API

Expected Outcome

The system should function as an intelligent research assistant capable of retrieving academic knowledge, summarizing research papers, generating insights, and helping users perform research tasks more efficiently.